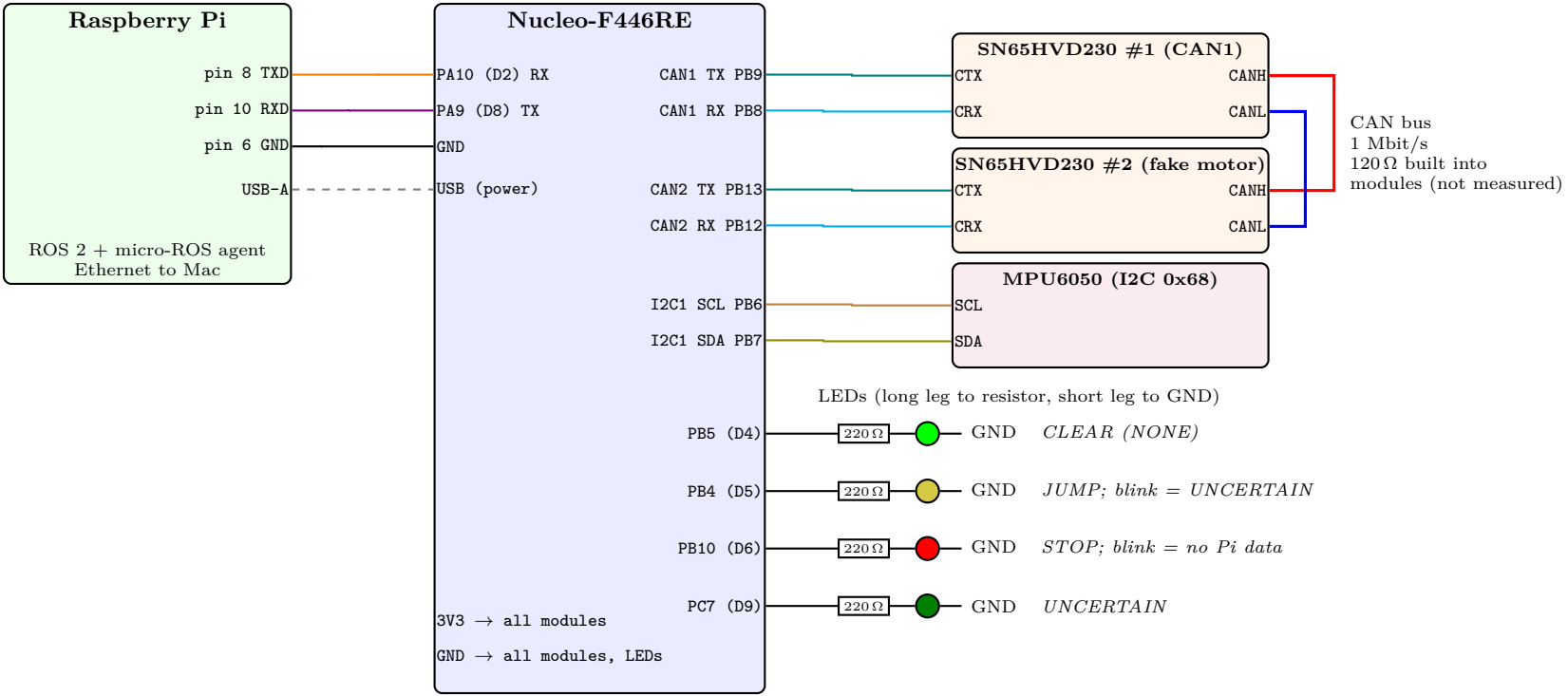
Raspberry Pi
pin 8 TXD
pin 10 RXD
pin 6 GND
USB-A
ROS 2 + micro-ROS agent
Ethernet to Mac
Nucleo-F446RE
PA10 (D2) RX
PA9 (D8) TX
GND
USB (power)
CAN1 TX PB9
CAN1 RX PB8
CAN2 TX PB13
CAN2 RX PB12
I2C1 SCL PB6
I2C1 SDA PB7
PB5 (D4)
PB4 (D5)
PB10 (D6)
PC7 (D9)
3V3 → all modules
GND → all modules, LEDs
SN65HVD230 #1 (CAN1)
CTX
CRX
CANH
CANL
SN65HVD230 #2 (fake motor)
CTX
CRX
CANH
CANL
MPU6050 (I2C 0x68)
SCL
SDA
CAN bus
1 Mbit/s
120 Ω built into
modules (not measured)
LEDs (long leg to resistor, short leg to GND)
220 Ω
GND
CLEAR (NONE)
220 Ω
GND
JUMP; blink = UNCERTAIN
220 Ω
GND
STOP; blink = no Pi data
220 Ω
GND
UNCERTAIN